

Directional Guidance Device Using Skin Shear Deformation on the Nasal Bridge

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ABSTRACT

Haptic feedback on the nose has gained attention as a method for presenting information that does not interfere with the visual or auditory senses, particularly in the context of navigation. Among these approaches, methods utilizing eyeglass nose pads are promising as they impose minimal additional burden on the user. However, existing methods mainly rely on vibration stimuli, which face challenges such as sensory adaptation, discomfort during continuous use, and reduced discriminability due to the propagation of vibrations. To address these issues, we propose a directional guidance method that induces skin shear deformation using rotating motors embedded in the nose pads to provide clearer directional cues. We developed a prototype and conducted an experiment to identify eight types of stimulus patterns. The results showed an overall accuracy of 72.8%. In particular, the directional recognition accuracy (classifying into Left, Right, or Both) reached an extremely high rate of 96.9%. On the other hand, response time analysis suggested that cognitive load increased under complex conditions where both motors were active. Subjective evaluations highlighted that while the clarity of the stimuli was appreciated, the shifting of glasses caused by shear deformation posed a challenge for wearability. This study demonstrates the high discriminability of skin shear stimulation on the nasal region, but optimization of parameters balancing comfort and accuracy is necessary for practical application.

Index Terms: Haptic Interface, Skin Shear Deformation, Nose, Smart Glasses

1 INTRODUCTION

Visual and auditory cues, commonly used in navigation, pose safety concerns by distracting users during movement. Consequently, haptics is expected to serve as a third modality. Numerous navigation methods using vibration motors to convey direction and distance via tactile patterns have been proposed. However, many such methods require wearing specialized devices on the wrist or waist, making attachment and removal cumbersome. To resolve this, recent proposals have integrated navigation systems into daily accessories, such as eyeglasses. For instance, Nakamura et al. [3] proposed a method that applies vibration stimuli to the left and right sides of the nasal bridge using nose pads. However, their study reported that prolonged vibration caused itchiness and burden on the user. Furthermore, applying vibration to a narrow area like the nose can make directional discrimination difficult due to vibration propagation.

In contrast, skin shear deformation, which applies tangential force to stretch the skin, has been proposed in haptics research to convey richer directional information than simple vibration patterns. Examples include methods that stretch facial skin [5] or apply friction to the nasal tip [2] to induce directional perception.

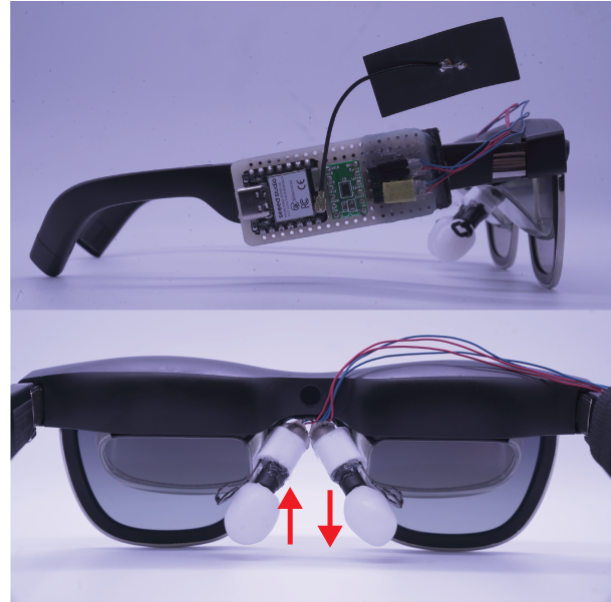


Figure 1: Device implementation with two rotors. The motors rotate to induce upward (or downward) skin shear deformation on the left (or right) side of the nose, as indicated by the arrows.

We hypothesize that skin shear deformation offers a fundamental solution to the comfort issues associated with vibration. Physiologically, conventional vibration (typically around 150 Hz) primarily activates rapidly adapting mechanoreceptors, specifically Pacinian corpuscles (FA-II) [1]. On the sensitive skin of the face, such high-frequency oscillation mimics the sensation of crawling insects or foreign objects, triggering defensive reactions perceived as “knismesis” (tickling) or itchiness [3]. In contrast, low-frequency skin shear deformation targets slowly adapting receptors, such as Ruffini endings (SA-II), which are specialized for detecting skin stretch and directional force [1]. Since this stimulation resembles natural physical contacts—such as the slight shifting of glasses—it is processed as non-threatening tactile information rather than a foreign irritant. This physiological distinction suggests that shear stimulation can provide clear directional cues with significantly reduced discomfort compared to vibration.

In this study, we propose a method to present direction by inducing skin shear deformation on the side of the nasal bridge using rotating motors in the nose pads. By leveraging the shear mechanism, we aim to achieve clearer directionality than vibration while avoiding issues of identification loss due to propagation and discomfort during long-term use. We developed a prototype and conducted an experiment to investigate the discriminability of these stimuli.



Figure 2: Response screen displayed during the experiment. Each arrow represents one of the 8 stimulus combinations. Participants responded using the corresponding keys on a numeric keypad.

2 PROPOSED METHOD

2.1 Implementation

The developed device (Fig. 1) consists of a nose pad unit with rotors and an electronic circuit unit. The nose pad unit features two gear motors (Pololu item: 2359, gear ratio 700:1, rated 6V, 90 rpm) equipped with rotors made of plastic clay (approx. 10 mm diameter). The circuit unit comprises a microcontroller (XIAO ESP32C6) for PC communication and motor control, and a motor driver (DRV8835). In this experiment, the microcontroller functioned as an access point, receiving signals from a PC via Wi-Fi (UDP) to control the rotation of the left and right motors for stimulus presentation.

2.2 Stimuli Design

The pressure sensitivity threshold of the nose is approximately 1.8 mm [6]. Therefore, we adopted a stimulus that deforms the skin by approximately 2 mm. The stimulus duration was set to 50 ms, with a motor rotation speed of approximately 4 cm/s. During control, different PWM duty cycles were applied to the left and right motors to calibrate for individual motor differences.

The basic stimulus pattern consisted of a 50 ms rotation in a specific direction (approx. 2 mm displacement), a 1000 ms pause, and a 50 ms return rotation. The stimulus state that can be produced by a single motor has three possible patterns:

1. The nasal skin is deformed upward and then returns to its original state.
2. The nasal skin is deformed downward and then returns to its original state.
3. No rotation is applied.

Excluding the condition in which neither motor rotates, the left-right motor combination yields 8 possible states. The 8 stimulus patterns are defined as follows:

1. UN: Left Up only (Left fwd→rev, Right stop)
2. DN: Left Down only (Left rev→fwd, Right stop)
3. NU: Right Up only (Left stop, Right rev→fwd)
4. ND: Right Down only (Left stop, Right fwd→rev)
5. UU: Both Up (Left fwd→rev, Right rev→fwd)
6. UD: Left Up / Right Down (Left fwd→rev, Right fwd→rev)
7. DU: Left Down / Right Up (Left rev→fwd, Right rev→fwd)
8. DD: Both Down (Left rev→fwd, Right fwd→rev)

This experiment evaluated whether participants could discriminate among these 8 stimulus patterns.

3 EXPERIMENT

The purpose of the experiment was to verify the discriminability of the rotational direction combinations presented by the proposed method. The study was approved by the Institutional Review Board of The University of Tokyo (Approval No. 24-59).

Fig. 2 shows the experimental setup. After completing a pre-experiment questionnaire, participants sat on a chair wearing the device. Eight types of stimuli were presented randomly. Participants identified the presented stimulus using a numeric keypad. The 8 patterns were mapped to specific keys, and participants were instructed to judge the direction of the initial movement and press the corresponding key. The next stimulus was presented 1 second after a response. One session consisted of 40 trials (5 repetitions × 8 types). We conducted 3 sessions in total: 1 practice session and 2 main sessions. Data from the 2 main sessions were used for analysis. Before the main experiment, participants were briefed on the mapping between the vertical skin deformation and the horizontal directional cues (e.g., vertical shear on the left pad corresponds to a "Left" cue). All participants demonstrated a clear understanding of this mapping during the practice session.

The system automatically recorded responses, correctness, and response times. Response time was defined as the duration from the onset of the stimulus presentation to the moment the participant pressed a key. Participants were instructed to wear the device such that both contact pads firmly touched the nasal skin. If the device shifted during the experiment, they were asked to reposition it to ensure consistent contact. After the experiment, we evaluated workload using NASA-TLX, usability using the System Usability Scale (SUS), and subjective impressions using a 7-point Likert scale and open-ended questions. The custom questionnaire covered fit (e.g., "Field of view was not obstructed," "Device fit was stable") and stimulus quality (e.g., "Intensity was appropriate," "Not unpleasant," "Motor sound was not distracting," "Vibration did not propagate to other parts," "Effective for target search without audio-visual cues," "Safe for use while walking," "Could clearly distinguish Left/Right/Front").

4 RESULTS

Six male participants (aged 21-28) took part in the experiment. Three of them were daily spectacle wearers. We analyzed data from the main sessions (720 total trials; 480 trials after excluding practice). Response time outliers (less than 0.2 s or more than 10.0 s) were removed (2 trials, 0.42%), resulting in 478 valid trials.

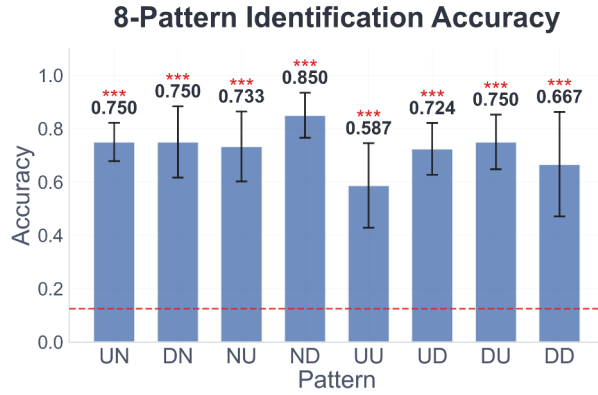


Figure 3: Accuracy by directional category (Left, Right, Both) and binomial test results against chance level (red dashed line). (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.005$)

4.1 Accuracy Analysis

Fig. 3 shows the accuracy for each of the 8 patterns. The overall accuracy was 72.8% (SD = 27.4%). A binomial test indicated that this accuracy was significantly higher than the chance level for 8 choices (12.5%) ($p < 0.000001$). The highest accuracy was observed for the ND (Right Down) pattern at 85.0%, while the lowest was for the UU (Both Up) pattern at 59.3%. Detailed analysis revealed that confusion mostly occurred between patterns with the same laterality (e.g., confusing UU with DD occurred in approx. 5% of cases), whereas confusion between different lateralities (Left, Right, Both) was extremely rare.

4.2 Directional Accuracy

While the exact identification of 8 patterns yielded 72.8% accuracy, the discrimination of the three directional categories (Left, Right, Both) was extremely high. Here, "directional recognition" is defined as correctly identifying the active side(s). For example, if a "Left Up" (UN) stimulus was identified as "Left Down" (DN), it is counted as a correct directional response (Left group). Similarly, correct classification into the Right group (NU, ND) or Both group (UU, UD, DU, DD) was evaluated.

Fig. 4 shows the accuracy based on these directional categories. The overall directional accuracy reached 96.9%, which was significantly higher than the chance level for 3 choices (33.3%) ($p < 0.000001$).

4.3 Response Time

Table 1 shows the response times by stimulus type.

Stimulus	Mean (ms)	SD (ms)
UN (Left Up)	2566	536
DN (Left Down)	2313	578
NU (Right Up)	2603	521
ND (Right Down)	2406	843
UU (Both Up)	2619	609
UD (Left Up / Right Down)	3059	854
DU (Left Down / Right Up)	3065	1035
DD (Both Down)	2618	1221

When grouped by direction:

- Left group: Mean 2473 ms (SD = 447 ms)

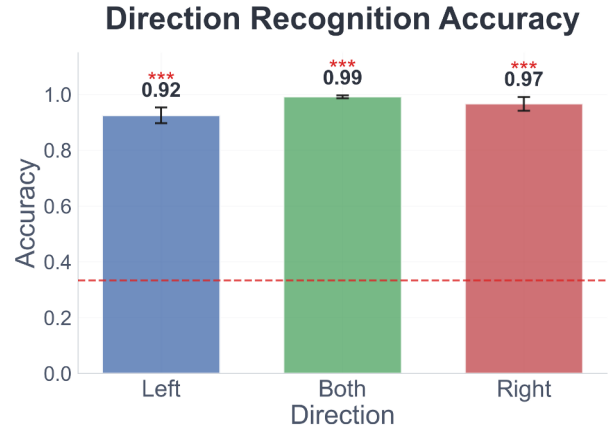


Figure 4: Accuracy by directional category (Left, Right, Both) and binomial test results against chance level (red dashed line). (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.005$)

- Right group: Mean 2492 ms (SD = 598 ms)

- Both group: Mean 2785 ms (SD = 917 ms)

T-tests revealed:

- Left vs. Right: No significant difference ($p = 0.883$)
- Left vs. Both: Significant difference ($p = 0.006$)
- Right vs. Both: Significant difference ($p = 0.032$)

These results indicate that processing stimuli involving both sides requires significantly more time than single-side stimuli, suggesting a higher cognitive load.

4.4 Questionnaire Results

4.4.1 Workload (NASA-TLX)

We evaluated subjective workload using the NASA-TLX (0-100 scale). Results are shown in Table 2.

Table 2: NASA-TLX Workload Evaluation

Subscale	Mean	SD
Mental Demand	68.0	17.2
Effort	46.0	29.7
Performance	38.0	22.5
Frustration	31.0	24.8
Temporal Demand	29.0	14.7
Physical Demand	23.0	11.5

Mental Demand (68.0) showed the highest workload, confirming that the identification of 8 types of tactile patterns is a cognitively difficult task. Next, Effort (46.0) was high, indicating that considerable concentration was required to accomplish the task. On the other hand, Physical Demand (23.0) was the lowest, confirming that the physical burden caused by the light contact stimulation to the nose was minimal. Also, Temporal Demand (29.0) was relatively low, suggesting that participants evaluated that they could work on the task at their own pace. Self-evaluation of Performance (38.0) showed a moderate value, suggesting that participants properly perceived their own identification performance.

4.4.2 Usability (SUS)

We conducted an objective usability evaluation using the System Usability Scale (SUS). SUS is evaluated in a range of 0 to 100, and generally, 68 or higher is the standard considered to be "good." The results of the SUS evaluation by 6 participants are shown in Table 3.

Table 3: SUS (System Usability Scale) Results

Item	Result
Mean Score	59.17
SD	11.80
Max Score	75.0
Min Score	42.5

The average SUS score obtained in this study (59.17) resulted in a value 8.83 points lower than the standard value generally considered "good" (68). This indicates that while it is a reasonable level for a device at the prototype stage, improvement of usability is necessary for practical application. Also, from the standard deviation (11.80), it was confirmed that there are certain individual differences in the perception of usability among participants. This variation is considered to be attributed to differences in individual sensitivity regarding fit and operability.

4.4.3 Fit, Comfort, and Clarity (Custom Questionnaire)

In our custom questionnaire, we administered eight questions and an open-ended section mainly focusing on wearability and the comfort and clarity of the presented stimuli. Regarding wearability, opinions were divided among participants. In the open-ended responses, multiple participants commented that "I felt the glasses slipping down during upward rotation, so I frequently adjusted their position." It is considered that the large amount of shear deformation caused the glasses to shift, thereby reducing wearability. Since this issue with wearability is considered to have a significant negative impact on comfort as well, it represents a challenge for the current stimuli.

On the other hand, regarding the comfort and clarity of the stimuli, multiple responses were received such as, "Surprisingly, once I got used to it, I could recognize it instantly, and since the nasal stimulation wasn't too intense, there was no discomfort." This suggests that the stimuli were easy to understand.

5 DISCUSSION

The experimental results demonstrated that the eight types of skin shear deformation patterns are distinguishable. In particular, the accuracy for "directional recognition"—identifying whether the active rotor was on the left, right, or both sides—reached an extremely high rate of 96.9%. This finding supports the validity of skin shear deformation as an effective modality for directional guidance. However, recognizing complex patterns where the left and right rotors moved differently imposed a higher cognitive load. Additionally, subjective feedback indicated that the physical shifting of the glasses caused by the shear force led to discomfort. This suggests that while shear deformation avoids the physiological irritation of vibration (e.g., itchiness), the mechanical stability of the eyewear is a critical factor for usability. To address this, simply reducing the stimulus magnitude could improve comfort but might degrade accuracy. Previous research suggests that deformation speed also affects directional perception [2]. Therefore, future work should identify the optimal balance between displacement magnitude and rotation speed to maintain high accuracy while minimizing device shifting. Furthermore, regarding the device form factor, the choice of the nasal bridge offers distinct cognitive advantages over other potential locations, such as the wrist. Tactile perception on the

hand often requires complex coordinate transformations depending on the hand's orientation relative to the body [4]. In contrast, stimulation on the face remains aligned with the user's egocentric forward direction. This theoretical advantage aligns with our qualitative findings; participants reported that the directional cues were easy to understand and could be recognized instantly once they acclimated to the sensation. This suggests that the nasal location minimizes the cognitive load required to interpret directional cues during navigation compared to limb-based systems.

5.1 Limitations

First, the current study was limited to a small sample of young male participants. Since tactile sensitivity and facial fit can vary by gender, age, and face shape, future evaluations must include a more diverse demographic. Second, although the motor sound was minimal, we did not use noise-canceling headphones; thus, we cannot strictly rule out the influence of faint auditory cues on discrimination performance.

6 CONCLUSION

In this study, we proposed a directional guidance method using skin shear deformation on the nasal bridge and clarified its high discriminability. Theoretical and physiological considerations suggest that this approach offers advantages in terms of spatial reference frames and reduced sensory adaptation compared to vibration or wrist-worn devices. Future steps will involve optimizing stimulus parameters to balance accuracy and comfort, and evaluating the method's effectiveness in practical navigation applications.

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